

The scheme — a prior on the geometry bridge, then alternating minimization

the prior supplies the information the data cannot; the alternation is what makes each unknown computable from the other

Training (offline, on simulated motion)

$$x_t = \text{FDK}(y, P_{nom}T(t\theta)) + t\Delta, \quad \Delta = x_{static} - \text{FDK}(y, P_{nom}T(\theta))$$

the geometry bridge: $t=0$ is the uncorrected recon (the inference cold start), $t=1$ the motion-free scan.

(9)

$$\psi^\star = \arg\min_{\psi} \mathbb{E}_{t,\theta,x} \|v_{\psi}(x_t,t) - \dot{x}_t\|^2, \quad \dot{x}_t = \frac{d}{dt}x_t$$

flow matching against the EXACT tangent of the bridge, so v_{ψ} points along the set of reachable reconstructions.

(10)

Inference (blind; $k=0,\dots,N-1$, $t_k=k/N$, $\Delta t=1/N$, $N=50$)

$$\tilde{x}_k = x_k + \Delta t \, v_{\psi^\star}(x_k, t_k)$$

PREDICT — the prior moves first, evaluated on 64^3 tiles and blended back to the volume.

(11)

$$\theta_{k+1} \approx \arg\min_{\theta} L(A_{P(\theta)}\tilde{x}_k, y)$$

ESTIMATE — 50 warm-started Adam steps on a coordinate net over the view index. Note the argument: $\tilde{x}_k$, not x_k (Gauss–Seidel).

(12)

$$z_k = \arg\min_z \|A_{P(\theta_{k+1})}z - y\|^2$$

DATA — 5 CG iterations warm-started at $\tilde{x}_k$; the constraint set itself has just been re-aimed by (12).

(13)

$$x_{k+1} = z_k + \kappa(\text{TV}(z_k) - z_k)$$

PnP corrector, $\kappa=0.3$ — an OOD-safe denoiser, unlike using $v_{\psi^\star}$ itself inside the splitting.

(14)

$$\bar{\theta} = \frac{1}{K} \sum_{j=N-K+1}^N \theta_j, \quad x_{out} = \text{FDK}(y, P_{nom}T(\bar{\theta}))$$

readout, $K=2$: the estimator's tail is a period-2 limit cycle, so the mean is taken rather than the last step.

(15)